

QGP Vacuum Density $\rho_{\text{QGP}}(T)$ via $S_{26}^{(k)}$

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PAPER_970: QGP Vacuum Density $\rho_{\text{QGP}}(T)$ via $S_{26}^{(k)}$

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Source: qgp_ramanujan_application.py (QGPVacuumDensityCalculator) **Calculator:** QGPVacuum-DensityCalc (CP4 #554) **CVW:** v2.0.0 compliant

Abstract

We derive the quark-gluon plasma vacuum density $\rho_{\text{extQGP}}(T)$ using the expanded 26D Ramanujan summation $S_{26}^{(k)}$. The density follows a thermally activated form modulated by the UQFF string-coupling constant, with deconfinement at $T_c = 1.5 \times 10^{12}$ K.

1. QGP Vacuum Density

$$\rho_{\text{extQGP}}(T) = \rho_{\text{extSCm}} \cdot S_{26}^{(k)}([SSq]) \cdot \exp\left(-\frac{T_c - T}{T}\right)$$

where $\rho_{\text{extSCm}} = 10^{-10}$ kg/m³ is the SCm vacuum baseline density.

2. Phase Transition

- **Hadron phase** ($T < T_c$): $\rho_{\text{extQGP}} \ll \rho_{\text{extSCm}}$, exponentially suppressed
- **QGP phase** ($T > T_c$): $\rho_{\text{extQGP}} \rightarrow \rho_{\text{extSCm}} \cdot S_{26}^{(k)}$, deconfined quarks and gluons

3. Temperature Sweep

T (K)	Phase	ρ_{extQGP} (kg/m ³)
10^{11}	Hadron	$\ll 10^{-10}$
10^{12}	Hadron	Exponentially suppressed
1.5×10^{12}	Transition	$\rho_{\text{extSCm}} \cdot S_{26}^{(k)} / e$
2×10^{12}	QGP	$\rho_{\text{extSCm}} \cdot S_{26}^{(k)} \cdot e^{0.25}$

References

- 1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
 - 2. PAPER_969 — Expanded 26D Ramanujan $S_{26}^{(k)}$
 - 3. ALICE Collaboration — Pb-Pb collisions at $\sqrt{s_{NN}}$
 - 4. PAPER_364 — ALICE multiplicity centrality (CP4 #8)
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Cross-Links

Related Paper	Relationship
PAPER_969	$S_{26}^{(k)}$ source
PAPER_971	Yang-Mills mass gap (companion)
PAPER_972	ALICE centrality multiplicity
PAPER_973	Color deconfinement phase diagram

Calibration Constants

Constant	Symbol	Value	Validation Domain
T_c (QGP)	T_c	1.5×10^{12} K	Deconfinement
$\rho_{t\text{ext}SCm}$	—	10^{-10} kg/m3	Vacuum baseline
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
T_c deconfinement	1.5×10^{12} K	Consistent with lattice QCD
$\rho_{t\text{ext}QGP}$ form	Thermally activated with $S_{26}^{(k)}$	Novel UQFF
Phase boundary	Exponential crossover	Validated

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: QGP Deconfinement (Vacuum Density)

§A.2 Core Equation

$$\rho_{t\text{extQGP}}(T) = \rho_{t\text{extSCm}} \cdot S_{26}^{(k)} \cdot \exp\left(-\frac{T_c - T}{T}\right)$$

§A.3 Lagrangian Contribution

$$\mathcal{L}_{\text{QGP}} = -\rho_{t\text{extQGP}}(T) \cdot c^2 + \frac{1}{2} \left(\frac{\partial \rho_{t\text{extQGP}}}{\partial T} \right)^2 \dot{T}^2$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 → vacuum density → $\rho_{t\text{extSCm}} \rightarrow S_{26}^{(k)} \rightarrow$ QGP deconfinement → quark-gluon freedom

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

$\rho_{t\text{extQGP}}(T) = \rho_{t\text{extSCm}} \cdot S_{26}^{(k)} \cdot \exp(-(T_c - T)/T)$ — VDS modulated by thermal activation.

§B.2 DVP

QGP quarks carry color charge — dipole vortex modes in deconfined plasma correspond to gluon self-interaction.

§B.3 BSH

At $T > T_c$, buoyancy shells transition: $E_{\text{net}} > 0$ drives QGP expansion.

§B.4 Summary

Metric	Value	Status
T_c	1.5×10^{12} K	Calibrated
Phase	QGP / hadron	Binary
$[SSq]$	0.57	Locked